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SUBJECT : Cyber & Information Security

Aim: Web Security with SSL/TLS

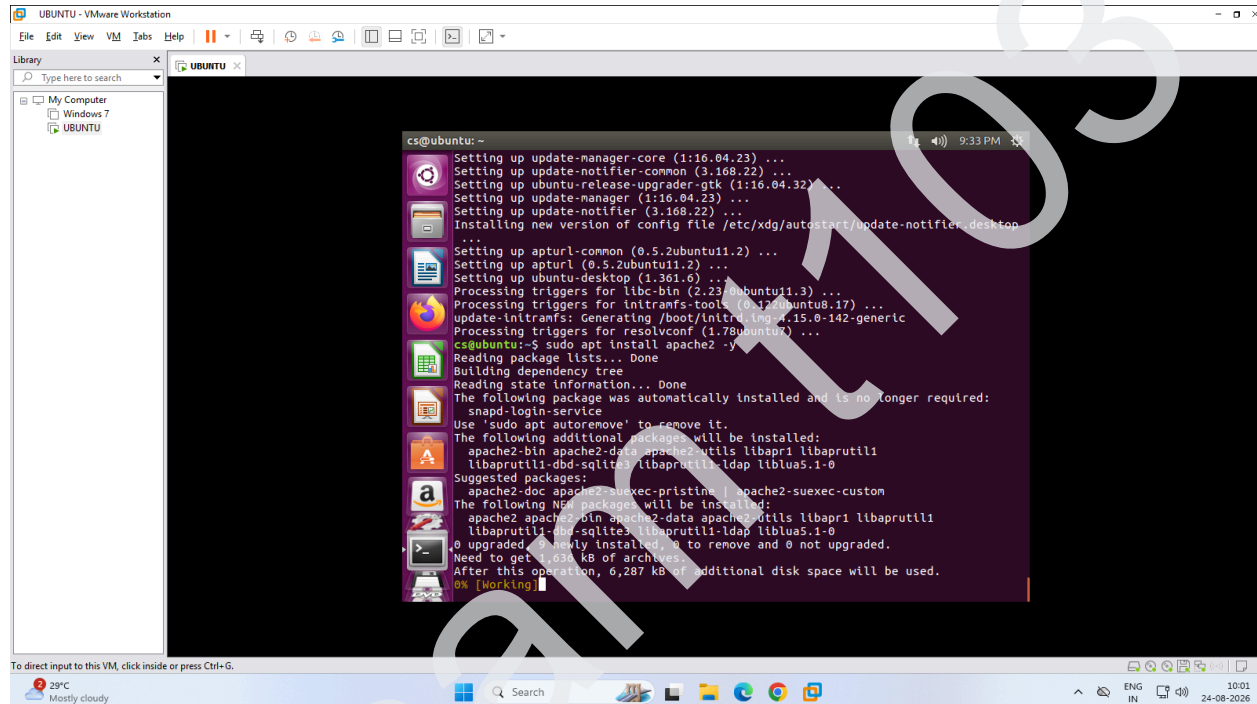
Step 1: Update the System Open Terminal and run:

```
sudo apt update
```

```
sudo apt upgrade -y
```

Step 2: Install Apache Web Server

```
sudo apt install apache2 -y
```

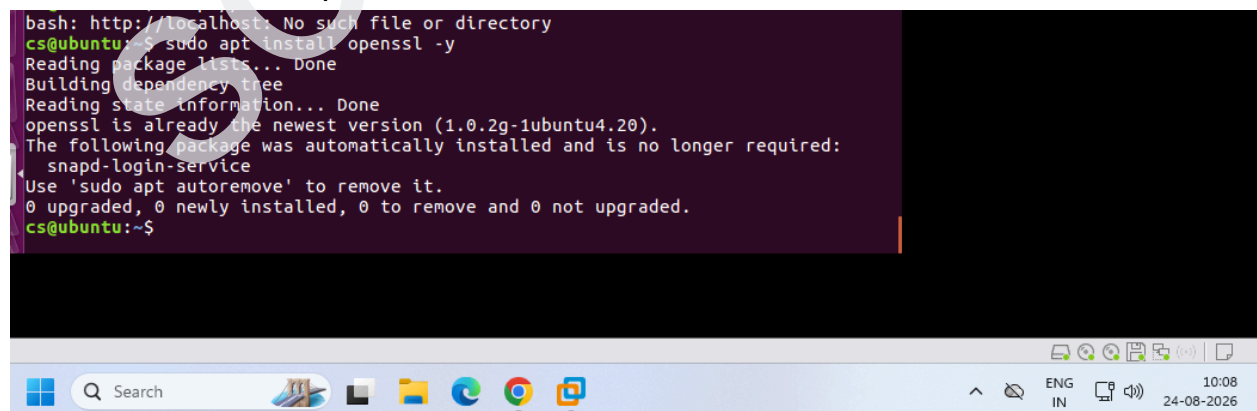


```
cs@ubuntu: ~$ sudo apt update
Setting up update-manager-core (1:16.04.23) ...
Setting up update-notifier-common (3.168.22) ...
Setting up ubuntu-release-upgrader-gtk (1:16.04.32) ...
Setting up update-manager (1:16.04.23) ...
Setting up update-notifier (3.168.22) ...
Installing new version of config file /etc/xdg/autostart/update-notifier.desktop
...
Setting up apturl-common (0.5.2ubuntu11.2) ...
Setting up apturl (0.5.2ubuntu11.2) ...
Setting up ubuntu-desktop (1.361.6) ...
Processing triggers for libc-bin (2.23-0ubuntu11.3) ...
Processing triggers for initramfs-tools (0.17ubuntu8.17) ...
update-initramfs: Generating /boot/initrd.img 4.15.0-142-generic
Processing triggers for resolvconf (1.78ubuntu7) ...
cs@ubuntu:~$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following package was automatically installed and is no longer required:
  snapd-login-service
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1 libaprutil1
  libaprutil1-dbd-sqlite3 libaprutil1-ldap liblua5.1-0
Suggested packages:
  apache2-doc apache2-suexec-pristine apache2-suexec-custom
  The following NP packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1 libaprutil1
  libaprutil1-dbd-sqlite3 libaprutil1-ldap liblua5.1-0
0 upgraded, 9 newly installed, 0 to remove and 0 not upgraded.
Need to get 1,638 kB of archives.
After this operation, 6,287 kB of additional disk space will be used.
0k Working
```

Step 3: Install OpenSSL OpenSSL is used to create and manage SSL/TLS certificates.

```
sudo apt install openssl -y
```

Check the version: openssl version



```
bash: http://localhost: No such file or directory
cs@ubuntu:~$ sudo apt install openssl -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
openssl is already the newest version (1.0.2g-1ubuntu4.20).
The following package was automatically installed and is no longer required:
  snapd-login-service
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
cs@ubuntu:~$
```

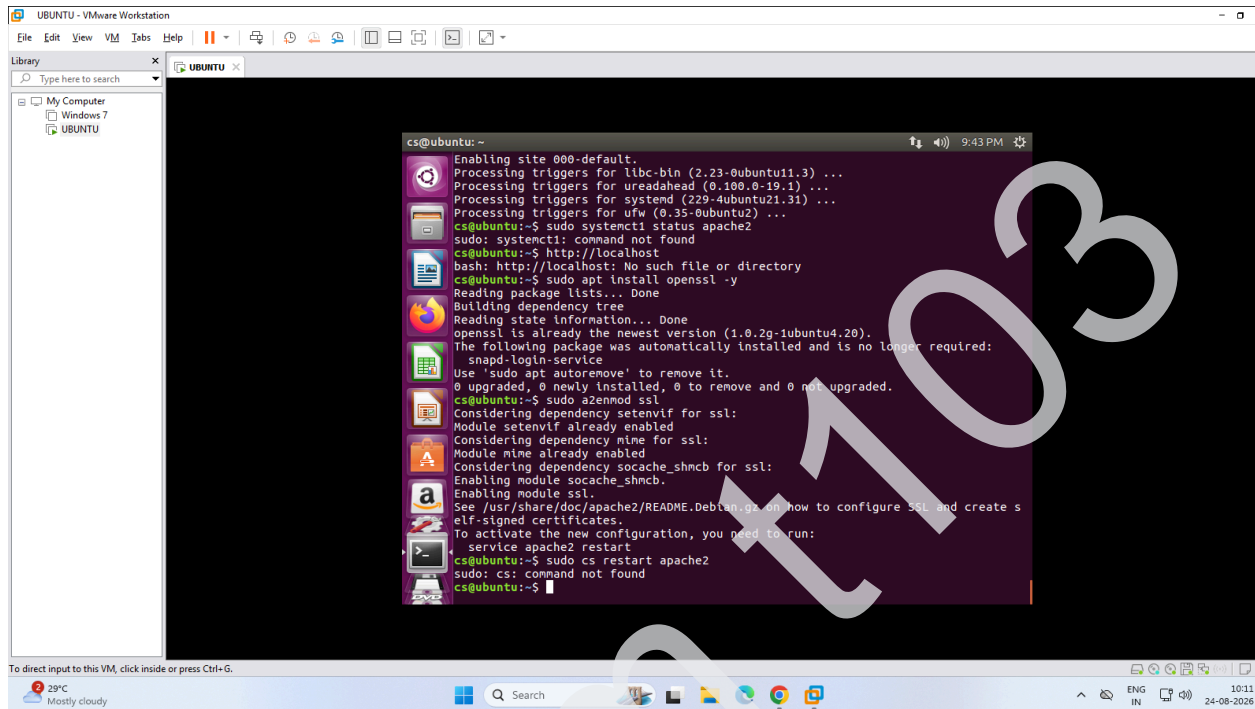
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Step 4: Enable SSL Module in Apache

Restart Apache:

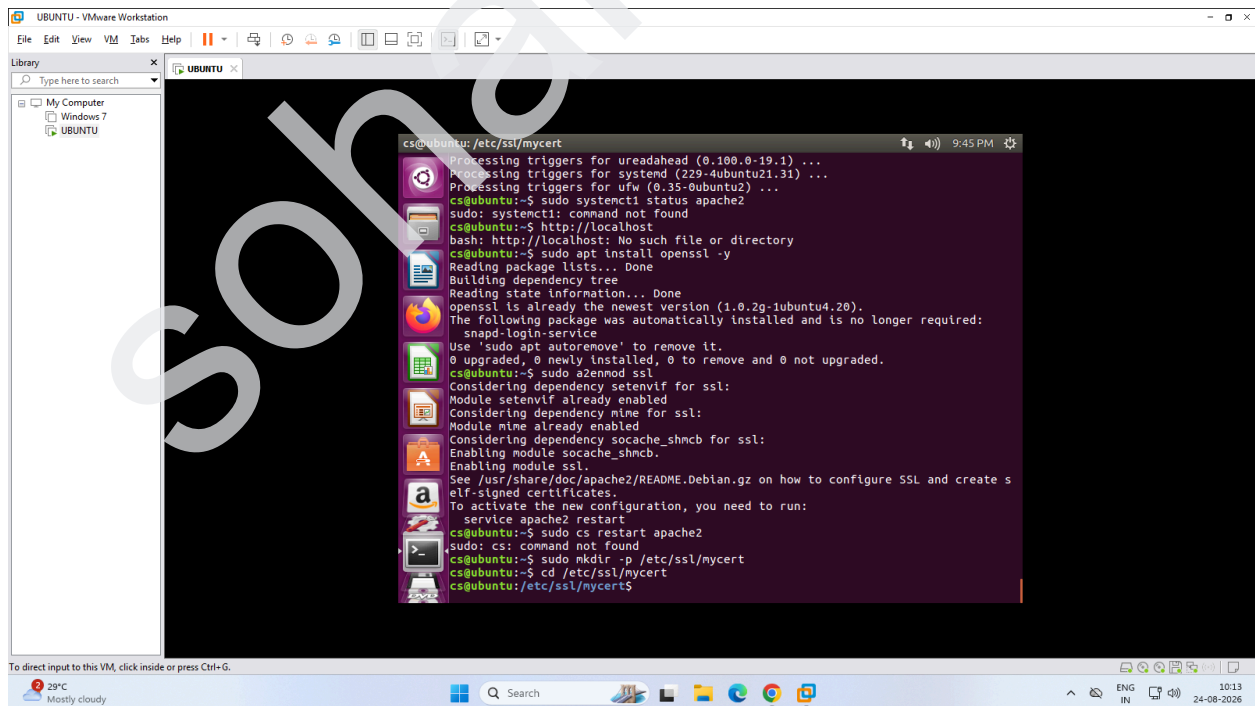
```
sudo systemctl restart apache2
```



```
cs@ubuntu:~$ sudo systemctl status apache2
sudo: systemctl: command not found
cs@ubuntu:~$ http://localhost
bash: http://localhost: No such file or directory
cs@ubuntu:~$ sudo apt install openssl -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
openssl is already the newest version (1.0.2g-1ubuntu4.20).
The following package was automatically installed and is no longer required:
  snapd-login-service
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
cs@ubuntu:~$ sudo a2enmod ssl
Considering dependency setenvif for ssl:
Module setenvif already enabled
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create s
elf-signed certificates.
To activate the new configuration, you need to run:
  service apache2 restart
cs@ubuntu:~$ sudo cs restart apache2
sudo: cs: command not found
cs@ubuntu:~$
```

Step 5: Create a Directory for Certificates

```
sudo mkdir -p /etc/ssl/mycert
```

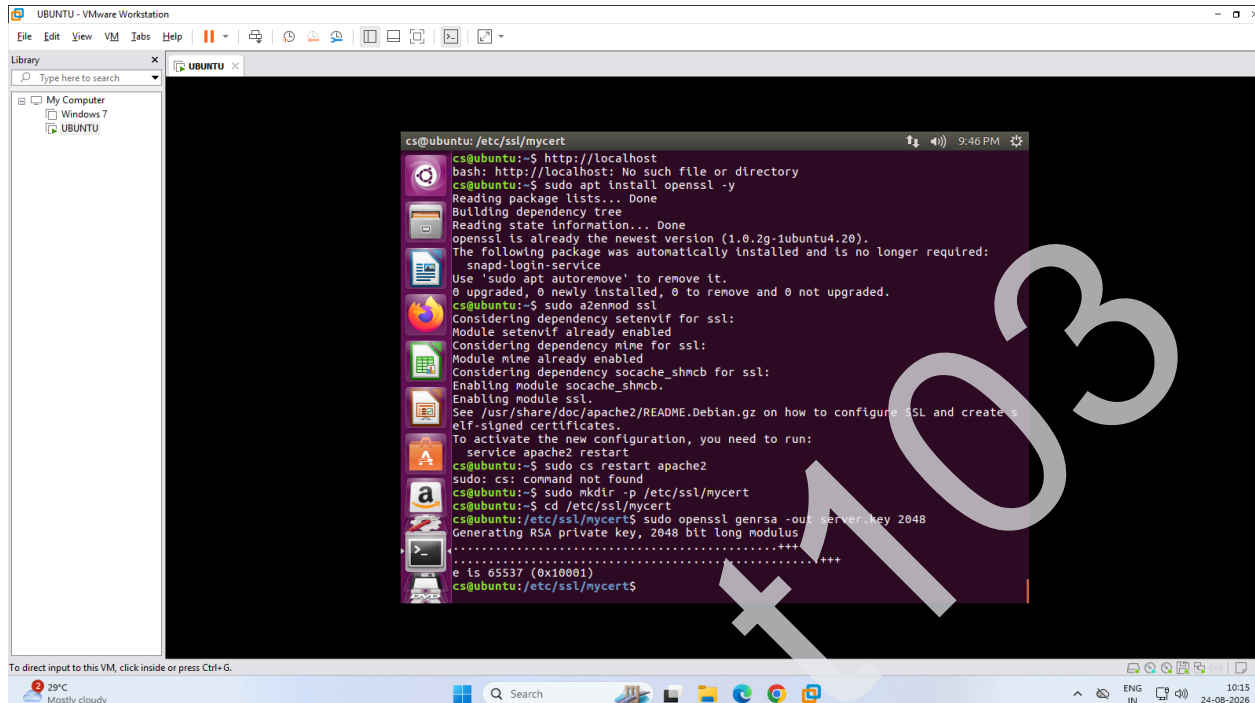


```
cs@ubuntu:/etc/ssl/mycert$
Processing triggers for ureadahead (0.100.0-19.1) ...
Processing triggers for systemd (229-4ubuntu21.31) ...
cs@ubuntu:~$ sudo systemctl status apache2
sudo: systemctl: command not found
cs@ubuntu:~$ http://localhost
bash: http://localhost: No such file or directory
cs@ubuntu:~$ sudo apt install openssl -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
openssl is already the newest version (1.0.2g-1ubuntu4.20).
The following package was automatically installed and is no longer required:
  snapd-login-service
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
cs@ubuntu:~$ sudo a2enmod ssl
Considering dependency setenvif for ssl:
Module setenvif already enabled
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create s
elf-signed certificates.
To activate the new configuration, you need to run:
  service apache2 restart
cs@ubuntu:~$ sudo cs restart apache2
sudo: cs: command not found
cs@ubuntu:~$ sudo mkdir -p /etc/ssl/mycert
cs@ubuntu:~$ cd /etc/ssl/mycert
cs@ubuntu:/etc/ssl/mycert$
```

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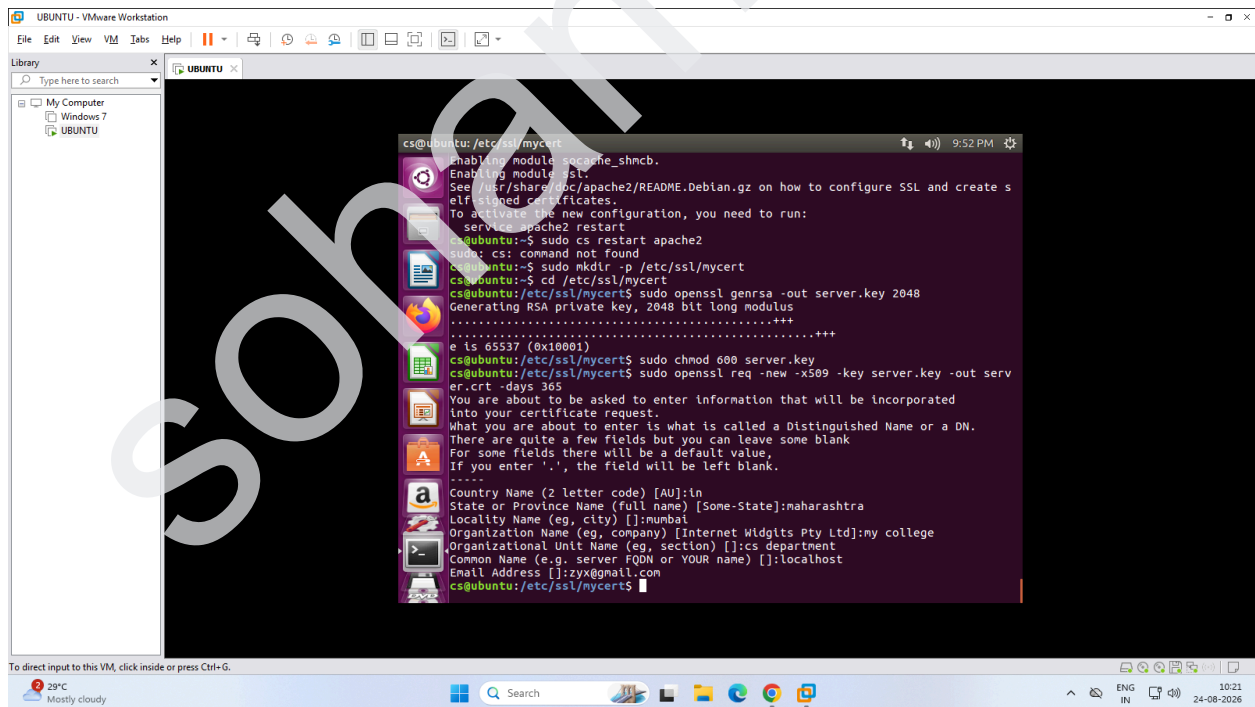
Step 6: Generate a Private Key



```
cs@ubuntu: /etc/ssl/mycert
cs@ubuntu:~$ http://localhost
bash: http://localhost: No such file or directory
cs@ubuntu:~$ sudo apt install openssl -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
openssl is already the newest version (1.0.2g-1ubuntu4.20).
The following package was automatically installed and is no longer required:
  snapd-login-service
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
cs@ubuntu:~$ sudo a2enmod ssl
Considering dependency setenvif for ssl:
Module setenvif already enabled
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create self-signed certificates.
To activate the new configuration, you need to run:
  service apache2 restart
cs@ubuntu:~$ sudo cs restart apache2
sudo: cs: command not found
cs@ubuntu:~$ sudo mkdir -p /etc/ssl/mycert
cs@ubuntu:~$ cd /etc/ssl/mycert
cs@ubuntu:/etc/ssl/mycert$ sudo openssl genrsa -out server.key 2048
Generating RSA private key, 2048 bit long modulus
.....+++++
e is 65537 (0x10001)
cs@ubuntu:/etc/ssl/mycert$
```

Step 7: Generate a Self-Signed SSL Certificate Run:

`sudo openssl req -new -x509 -key server.key -out server.crt -days 365`



```
cs@ubuntu: /etc/ssl/mycert
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create self-signed certificates.
To activate the new configuration, you need to run:
  service apache2 restart
cs@ubuntu:~$ sudo cs restart apache2
sudo: cs: command not found
cs@ubuntu:~$ sudo mkdir -p /etc/ssl/mycert
cs@ubuntu:~$ cd /etc/ssl/mycert
cs@ubuntu:/etc/ssl/mycert$ sudo openssl genrsa -out server.key 2048
Generating RSA private key, 2048 bit long modulus
.....+++++
e is 65537 (0x10001)
cs@ubuntu:/etc/ssl/mycert$ sudo chmod 600 server.key
cs@ubuntu:/etc/ssl/mycert$ sudo openssl req -new -x509 -key server.key -out server.crt -days 365
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
if you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:in
State or Province Name (full name) [Some-State]:maharashtra
Locality Name (eg, city) []:mumbai
Organization Name (eg, company) [Internet Widgits Pty Ltd]:my college
Organizational Unit Name (eg, section) []:cs department
Common Name (e.g. server FQDN or YOUR name) []:localhost
Email Address []:zyx@gmail.com
cs@ubuntu:/etc/ssl/mycert$
```

Step 8: Configure Apache for HTTPS

Enable the default SSL site:

`sudo a2ensite default-ssl`

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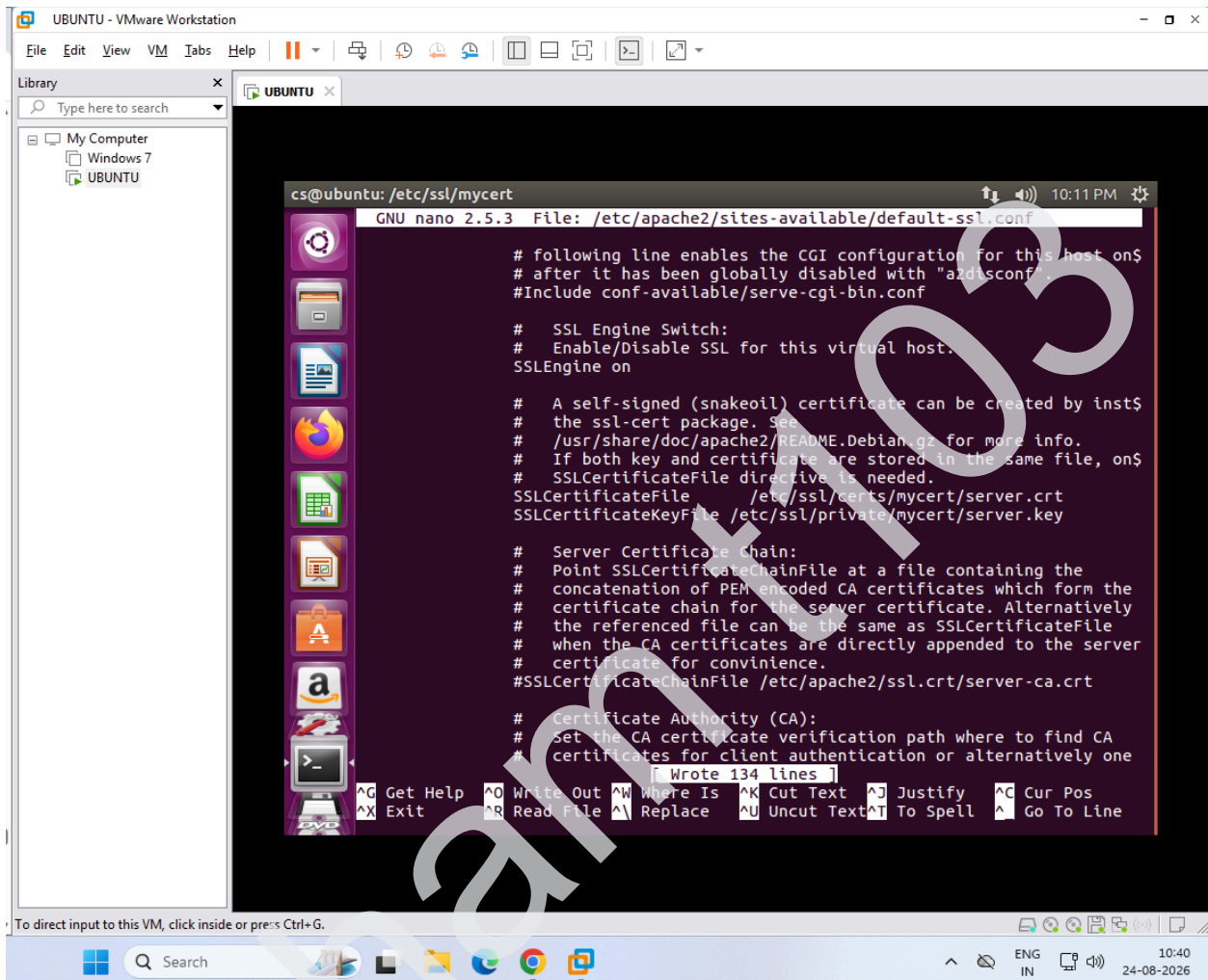
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Edit the SSL configuration:

`sudo nano /etc/apache2/sites-available/default-ssl.conf`



```
cs@ubuntu: /etc/ssl/mycert
GNU nano 2.5.3 File: /etc/apache2/sites-available/default-ssl.conf

# following line enables the CGI configuration for this host on$
# after it has been globally disabled with "a2disconf"
#Include conf-available/serve-cgi-bin.conf

#
# SSL Engine Switch:
# Enable/Disable SSL for this virtual host.
SSLEngine on

#
# A self-signed (snakeoil) certificate can be created by inst$
# the ssl-cert package. See
# /usr/share/doc/apache2/README.Debian.gz for more info.
# If both key and certificate are stored in the same file, on$
# SSLCertificateFile directive is needed.
SSLCertificateFile /etc/ssl/certs/mycert/server.crt
SSLCertificateKeyFile /etc/ssl/private/mycert/server.key

#
# Server Certificate Chain:
# Point SSLCertificateChainFile at a file containing the
# concatenation of PEM encoded CA certificates which form the
# certificate chain for the server certificate. Alternatively
# the referenced file can be the same as SSLCertificateFile
# when the CA certificates are directly appended to the server
# certificate for convinence.
#SSLCertificateChainFile /etc/apache2/ssl.crt/server-ca.crt

#
# Certificate Authority (CA):
# Set the CA certificate verification path where to find CA
# certificates for client authentication or alternatively one
# certificate for convinence.
Wrote 134 lines
^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos
^X Exit ^R Read File ^A Replace ^U Uncut Text ^T To Spell ^_ Go To Line
```

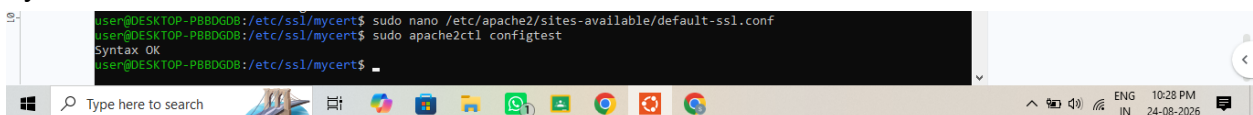
Step 9: Test Apache Configuration

Run:

`sudo apache2ctl configtest`

Expected result:

Syntax OK



```
user@DESKTOP-PBBGDB: /etc/ssl/mycert$ sudo nano /etc/apache2/sites-available/default-ssl.conf
user@DESKTOP-PBBGDB: /etc/ssl/mycert$ sudo apache2ctl configtest
Syntax OK
user@DESKTOP-PBBGDB: /etc/ssl/mycert$
```

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Step 10: Restart Apache

```
user@DESKTOP-PB8DGD8:/etc/ssl/mycert
HP09526: Syntax error on line 31 of /etc/apache2/sites-enabled/default-ssl.conf:
SSLCertificateFile: file '/etc/ssl/certs/mycert/server.crt' does not exist or is empty
user@DESKTOP-PB8DGD8:/etc/ssl/mycert$ sudo sed -i 's|/etc/ssl/certs/mycert|/etc/ssl/mycert|g' /etc/apache2/sites-available/default-ssl.conf
user@DESKTOP-PB8DGD8:/etc/ssl/mycert$ grep -n "SSLCertificate" /etc/apache2/sites-available/default-ssl.conf
10: # SSLCertificateFile directive is needed.
11: SSLCertificateFile /etc/ssl/mycert/server.crt
12: SSLCertificateKeyFile /etc/ssl/private/mycert/server.key
13: # Point SSLCertificateChainFile at a file containing the
14: # the referenced file can be the same as SSLCertificateFile
15: #SSLCertificateChainFile /etc/apache2/ssl.crt/server-ca.crt
user@DESKTOP-PB8DGD8:/etc/ssl/mycert$ sudo apache2ctl configtest
HP09526: Syntax error on line 31 of /etc/apache2/sites-enabled/default-ssl.conf:
SSLCertificateFile: file '/etc/ssl/mycert/server.crt' does not exist or is empty
user@DESKTOP-PB8DGD8:/etc/ssl/mycert$ sudo sed -i 's|SSLCertificateFile|SSLCertificateFile /etc/ssl/mycert/server.crt|; s|SSLCertificateKeyFile|SSLCertificateKeyFil
|' /etc/ssl/mycert/server.key' /etc/apache2/sites-available/default-ssl.conf
sed: can't read /etc/apache2/sites-available/default-ssl.conf: No such file or directory
user@DESKTOP-PB8DGD8:/etc/ssl/mycert$ ls -l /etc/apache2/sites-available/
total 12
-rw-r--r-- 1 root root 1286 Mar 18 2024 000-default.conf
-rw-r--r-- 1 root root 4558 Aug 24 16:43 default-ssl.conf
user@DESKTOP-PB8DGD8:/etc/ssl/mycert$ ls -l /etc/apache2/sites-enabled/
ls: cannot access '/etc/apache2/sites-enabled/': No such file or directory
user@DESKTOP-PB8DGD8:/etc/ssl/mycert$ ls -l /etc/apache2/sites-enabled/
total 0
-rwxrwxrwx 1 root root 35 Aug 24 14:56 000-default.conf -> ../sites-available/000-default.conf
-rwxrwxrwx 1 root root 35 Aug 24 16:20 default-ssl.conf -> ../sites-available/default-ssl.conf
user@DESKTOP-PB8DGD8:/etc/ssl/mycert$ sudo nano /etc/apache2/sites-available/default-ssl.conf
user@DESKTOP-PB8DGD8:/etc/ssl/mycert$ sudo apache2ctl configtest
syntax OK
user@DESKTOP-PB8DGD8:/etc/ssl/mycert$ sudo systemctl restart apache2
system has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
user@DESKTOP-PB8DGD8:/etc/ssl/mycert$ sudo service apache2 restart
user@DESKTOP-PB8DGD8:/etc/ssl/mycert$ sudo service apache2 status
* apache2 is running
user@DESKTOP-PB8DGD8:/etc/ssl/mycert$
```

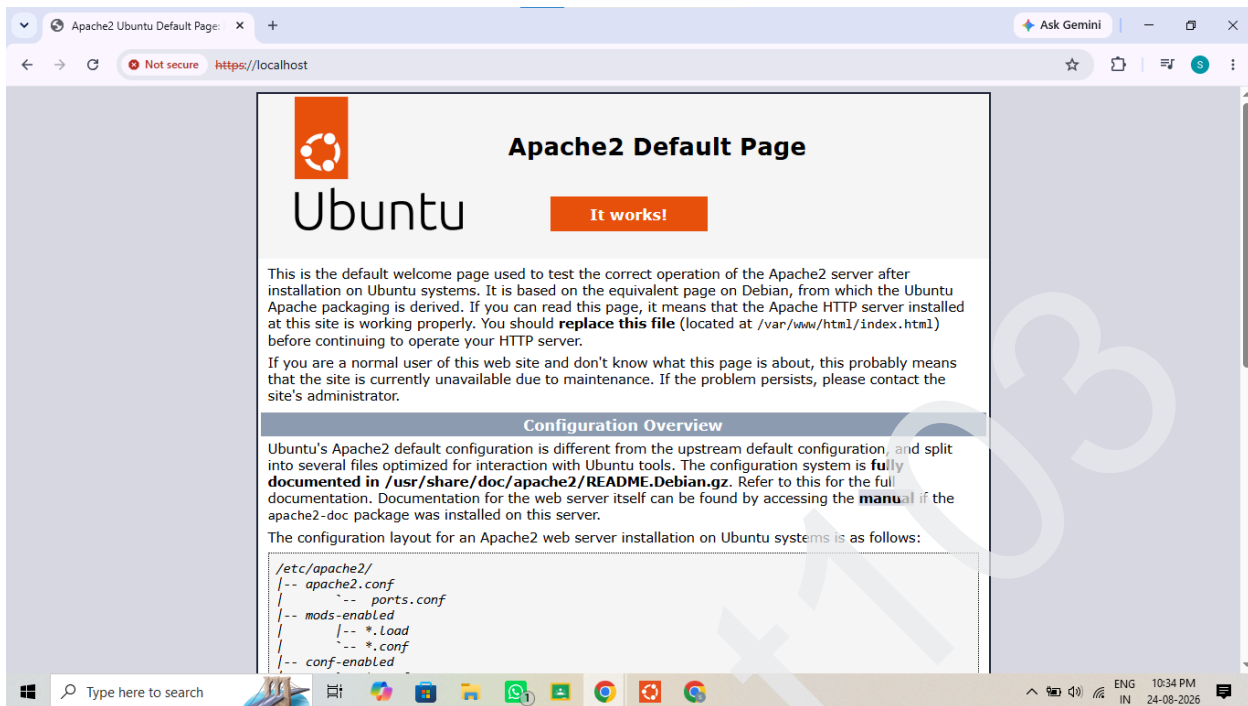
Step 11: Test Secure HTTPS Communication

Open a browser and visit:

<https://localhost>

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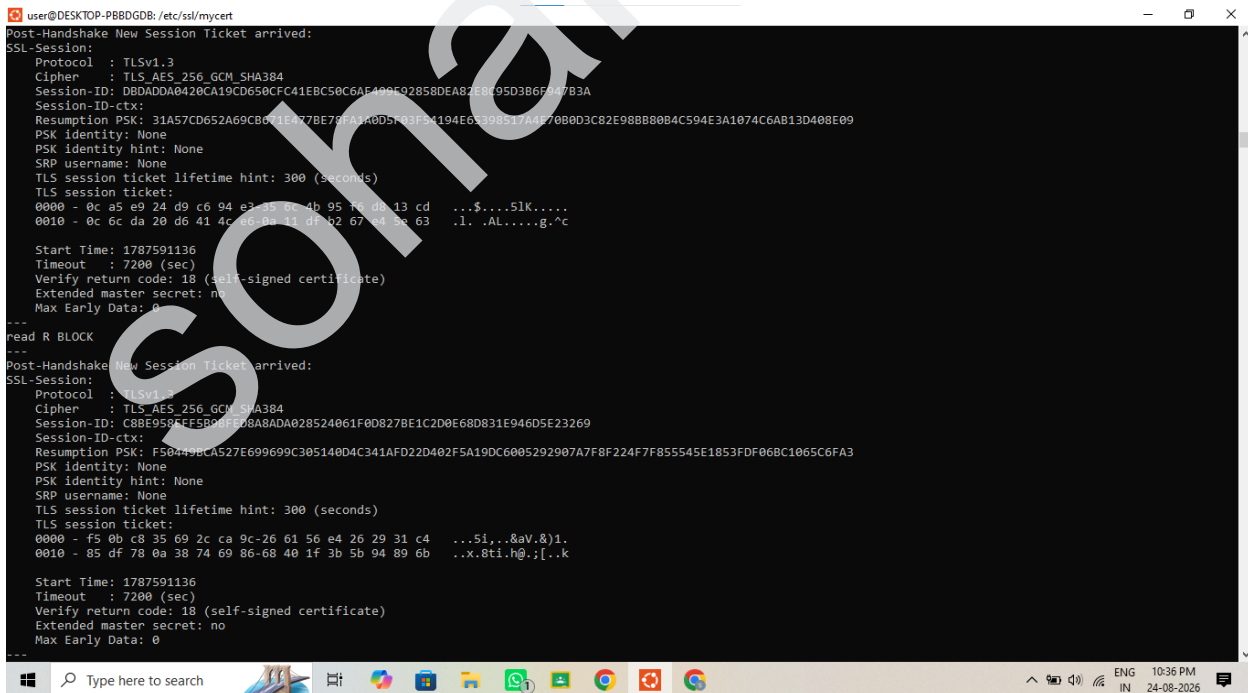
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Step 12: Verify the TLS Connection

Use OpenSSL to test the secure connection:

`openssl s_client -connect localhost:443`



Step 13: Check the Certificate

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Display certificate information:

openssl x509 -in /etc/ssl/mycert/server.crt -text -noout

```
user@DESKTOP-PBBDGDB: /etc/ssl/mycert
read R BLOCK
closed
user@DESKTOP-PBBDGDB:/etc/ssl/mycert$ openssl x509 -in /etc/ssl/mycert/server.crt -text -noout
Certificate:
  Data:
    Version: 3 (0x2)
    Serial Number:
      50:08:fd:92:15:e7:a8:7b:4b:ba:6f:5f:86:30:8f:57:07:c3:1a:1a
    Signature Algorithm: sha256WithRSAEncryption
    Issuer: C = IN, ST = Maharashtra, L = Mumbai, O = My college, OU = CS department, CN = localhost, emailAddress = yzx@gmail.com
    Validity
      Not Before: Aug 24 16:20:03 2026 GMT
      Not After : Aug 24 16:20:03 2027 GMT
    Subject: C = IN, ST = Maharashtra, L = Mumbai, O = My college, OU = CS department, CN = localhost, emailAddress = yzx@gmail.com
    Subject Public Key Info:
      Public Key Algorithm: rsaEncryption
      Public-Key: (2048 bit)
      Modulus:
        00:9d:22:99:61:e4:de:98:27:61:2c:3c:1c:55:7c:
        15:0a:6e:d6:80:52:8b:8d:56:19:47:56:3f:a4:c6:
        a0:4d:43:ba:37:41:08:bd:8e:3b:57:6d:cf:4e:0c:
        4f:63:09:4d:0a:a4:00:8f:e8:a5:ac:f0:dd:41:a4:
        91:0b:ed:cf:65:b9:70:5e:d5:be:fd:fb:b4:56:20:
        91:c8:40:6d:ee:50:19:45:b0:45:6b:6d:04:75:b2:
        96:74:44:6c:26:9b:ae:dd:19:0c:c2:0a:22:91:3a:
        f4:16:93:17:6a:84:87:27:a6:e1:ae:da:5b:16:39:
        85:a9:bc:ea:4a:3c:0c:b7:c7:b0:14:e9:ac:fc:50:
        c9:86:6c:9e:16:44:c2:dc:4e:f4:f8:cb:72:39:
        b7:fc:76:3f:32:60:a9:b8:59:a3:a7:09:f3:ed:58:
        63:f2:bb:e4:c6:25:30:d4:03:b5:38:42:fb:70:bc:
        24:3a:fd:e1:09:86:04:dc:2a:0a:7f:ee:b3:63:ce:
        b5:2f:05:c1:85:02:63:f1:9f:fa:7a:2f:45:cf:97:
        44:1b:69:ce:8b:a6:38:e5:b5:f3:b9:cd:76:1f:22:
        09:e2:a3:8a:76:19:ce:19:4f:25:af:70:87:59:d5:
        b1:e3:ac:03:90:10:5d:80:98:b6:4b:e4:5b:25:a7:
        f2:89
      Exponent: 65537 (0x10001)
    X509v3 extensions:
      X509v3 Subject Key Identifier:
        A0:0C:52:BA:E7:AC:53:C3:CC:F1:98:38:21:7D:45:C3:84:19:29:4C
      X509v3 Authority Key Identifier:
        A0:0C:52:BA:E7:AC:53:C3:CC:F1:98:38:21:7D:45:C3:84:19:29:4C
      X509v3 Basic Constraints: critical
        CA:TRUE
      Signature Algorithm: sha256WithRSAEncryption
      Signature Value:
        95:8c:f2:36:62:df:d8:51:f1:39:e7:d0:e0:17:5c:61:76:bf:
        22:41:9b:24:23:d7:32:32:0a:93:e3:24:b4:19:95:9c:e5:53:
        1e:6c:1d:aa:1d:0c:8b:f1:0a:1f:97:56:09:ce:60:a0:16:04:
        f9:00:42:76:1e:0a:f5:a2:0b:bc:12:88:00:7e:54:44:26:90:
        1b:78:81:1a:07:07:5d:f1:0a:23:0b:81:82:f2:e0:f3:1a:
        57:ed:41:0a:e2:9f:8e:b8:80:4a:09:16:36:ee:e8:20:fd:aa:
        9e:31:4e:8c:89:f2:27:6a:da:d5:4a:77:2b:c0:37:7a:b4:cb:
        38:e0:03:21:62:ed:7e:01:0:aa:ef:e8:81:81:1d:04:63:d5:
        3a:85:04:0a:0e:00:2f:4f:17:31:50:3c:4a:10:f0:28:4b:af:
        dd:91:02:00:f1:cc:38:ec:05:0e:b8:80:b1:bd:2e:cb:98:59:
        cd:b9:94:33:3b:a2:28:42:b0:b0:bf:d8:89:91:26:35:7f:6e:
        db:38:13:6c:9d:b5:df:9a:cd:27:6f:bc:ad:0c:d6:bc:d7:23:
        8e:b9:d0:54:a2:02:0e:bc:f0:a2:91:b5:0e:00:c0:f0:dc:
        1c:3e:3b:7a:bc:00:45:ce:77:c0:8f:d6:dd:02:5e:55:87:60:
        17:14:fe:a5
user@DESKTOP-PBBDGDB:/etc/ssl/mycert$
```

Step 14: Secure Certificate Management

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Check certificate expiry date:

`openssl x509 -enddate -noout -in /etc/ssl/mycert/server.crt`

```
user@DESKTOP-PBBDGDB: /etc/ssl/mycert
c0:86:6c:9e:16:44:c2:dc:4e:f4:f8:f8:cb:72:39:
b7:fc:76:3f:32:60:a0:b8:59:a3:a7:00:f3:ed:58:
63:f2:bb:e4:c6:25:30:d4:03:b5:38:42:fb:70:bc:
24:3a:fd:e1:09:86:04:dc:2a:0a:7f:ee:b3:63:ce:
b5:2f:05:c1:85:02:63:f1:9f:fa:7a:2f:45:cf:97:
44:1b:69:ce:8b:a6:38:e5:b5:f3:b9:cd:76:1f:22:
09:e2:a3:8a:76:19:ce:19:4f:25:af:70:87:59:d5:
b1:e3:ac:03:90:10:5d:80:98:b6:4b:e4:5b:25:a7:
f2:89
Exponents: 65537 (0x10001)
X509v3 extensions:
X509v3 Subject Key Identifier:
A0:0C:52:BA:E7:AC:53:C3:CC:F1:98:38:21:7D:45:C3:84:19:29:4C
X509v3 Authority Key Identifier:
A0:0C:52:BA:E7:AC:53:C3:CC:F1:98:38:21:7D:45:C3:84:19:29:4C
X509v3 Basic Constraints: critical
CA:TRUE
Signature Algorithm: sha256WithRSAEncryption
Signature Value:
95:8c:f2:36:62:df:d8:51:f1:39:e7:d0:e0:17:5c:61:76:bf:
22:41:9b:24:23:d7:32:62:9a:93:e3:24:b4:f3:95:9c:e5:53:
1e:6c:1d:aa:1d:0c:0b:3f:24:1f:97:56:02:ce:60:a0:f6:04:
f9:00:42:76:1b:49:f5:a2:eb:bc:12:88:90:7e:54:44:26:90:
1b:78:81:f9:a7:f6:5d:5d:cf:54:63:9b:8b:82:f2:e0:f3:1a:
57:ed:4c:6d:e2:9f:8e:b8:00:f8:09:16:36:ee:e8:20:fd:aa:
9e:31:4e:8c:89:f2:27:68:4a:d5:4a:77:2b:c0:37:7a:b4:cb:
38:e6:03:21:62:ad:7e:61:92:aa:ef:e8:81:81:d1:04:63:d5:
3a:85:c4:06:6e:96:2f:4f:17:31:50:3c:4a:10:f0:28:4b:af:
dd:91:02:08:ff:cc:38:ec:95:0e:b8:80:b1:bd:2e:cb:98:59:
cd:b9:94:33:3b:a2:28:42:bb:b0:bf:d8:89:91:26:35:7f:6e:
db:38:13:6c:9d:96:df:94:cd:27:6f:bc:ad:0c:d6:bc:d7:23:
8e:b9:d0:54:a3:d2:9c:bc:f0:a2:51:b5:6e:80:c0:f0:dc:
1c:3e:2b:7a:bc:60:45:ce:77:c0:8f:d6:dd:02:5e:55:87:60:
17:14:fe:a5
user@DESKTOP-PBBDGDB:/etc/ssl/mycert$ openssl x509 -enddate -noout -in /etc/ssl/mycert/server.crt
notAfter=Aug 24 16:20:03 2027 GMT
user@DESKTOP-PBBDGDB:/etc/ssl/mycert$
```

Step 16: Configure Strong TLS Security

Edit the SSL configuration:

`sudo nano /etc/apache2/mods-available/ssl.conf`

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```
user@DESKTOP-P8BDGDB: /etc/ssl/mycert
GNU nano 7.2 /etc/apache2/mods-available/ssl.conf
# Clients may prefer lower grade encryption. You should enable this
# option if you want to enforce stronger encryption, and can afford
# the CPU cost, and did not override SSLCipherSuite in a way that puts
# insecure ciphers first.
# Default: Off
#SSLHonorCipherOrder on

# The protocols to enable.
# Available values: all, SSLv3, TLSv1, TLSv1.1, TLSv1.2
# SSL v2 is no longer supported
SSLProtocol -all +TLSv1.2 +TLSv1.3

# Allow insecure renegotiation with clients which do not yet support the
# secure renegotiation protocol. Default: Off
#SSLInsecureRenegotiation on

# Whether to forbid non-SNI clients to access name based virtual hosts.
# Default: Off
#SSLStrictSNIVHostCheck On

# Warning: Session Tickets require regular reloading of the server!
# Make sure you do this (e.g. via logrotate) before changing this setting!
SSLSessionTickets off

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^J Execute    ^C Location  ^U Undo       ^M-A Set Mark  ^] To Bracket  ^M-O Previous
^X Exit      ^R Read File  ^N Replace    ^P Paste      ^O Justify    ^V Go To Line ^L Redo      ^M-G Copy     ^_ Where Was  ^M-N Next
```

Restart Apache:

sudo systemctl restart apache2

```
user@DESKTOP-P8BDGDB: /etc/ssl/mycert
b7:fc:76:3f:32:60:a9:b8:59:a3:a7:09:f3:ed:58:
63:f2:bb:e4:c6:25:30:d4:03:b5:38:42:fb:70:bc:
24:3a:fd:e1:09:86:04:dc:2a:0a:7f:ee:b3:63:ce:
b5:2f:05:c1:85:02:63:f1:9f:fa:7a:2f:45:cf:97:
44:1b:69:ce:8b:a6:38:e5:b5:f3:b9:cd:76:1f:f2:
09:e2:a3:8a:76:19:ce:19:4f:25:af:70:87:5d:d1:
b1:e3:ac:03:90:10:5d:80:98:b0:4b:e4:5b:25:a7:
f2:89
Exponent: 65537 (0x10001)
X509v3 extensions:
X509v3 Subject Key Identifier:
  A0:0C:52:BA:E7:AC:53:C3:CC:F1:98:38:21:7D:45:C3:84:19:29:4C
X509v3 Authority Key Identifier:
  A0:0C:52:BA:E7:AC:53:C3:CC:F1:98:38:21:7D:45:C3:84:19:29:4C
X509v3 Basic Constraints: critical
  CA:TRUE
Signature Algorithm: sha256WithRSAEncryption
Signature Value:
  95:8c:f2:36:62:df:d8:51:f1:30:e7:d0:e0:17:3e:61:76:bf:
  22:41:9b:24:23:d7:32:62:0d:03:03:02:b4:f3:99:9c:e5:53:
  1e:6c:1d:aa:1d:0c:8b:30:04:1f:97:c6:00:ce:0a:0b:f6:04:
  f9:00:42:76:1b:49:f5:12:eb:bc:12:83:00:7e:54:44:26:90:
  1b:78:81:f9:a7:f6:5d:54:cf:54:63:9b:0b:02:f2:e0:f3:1a:
  57:ed:4c:6d:e2:9f:8e:00:80:f0:09:16:35:0a:e8:20:fd:aa:
  9e:31:4e:8c:89:f2:27:08:a5:d5:4a:77:2b:0b:37:7a:b4:cb:
  38:e6:03:21:62:ad:7e:01:02:aa:ef:e8:81:11:d0:46:d5:
  3a:85:c4:00:6e:06:2f:4f:a7:13:50:3c:4a:10:f0:28:4b:af:
  dd:91:02:00:ff:cc:38:ec:99:0e:b8:80:b1:bd:2e:cb:98:59:
  cd:b9:94:13:3b:a2:28:49:bb:b0:bf:d8:89:91:26:35:7f:6e:
  db:38:13:6c:9d:06:df:94:03:27:6f:bc:ad:0c:d6:bc:d7:23:
  8e:b9:00:54:05:d2:0c:bc:f0:a2:51:b5:6e:80:c0:f0:dc:
  1c:3e:33:00:bb:00:45:ce:77:c0:8f:d6:dd:02:5e:55:87:60:
  17:14:fe:95
user@DESKTOP-P8BDGDB:/etc/ssl/mycert$ openssl x509 -enddate -noout -in /etc/ssl/mycert/server.crt
notAfter=Aug 24 16:20:48 2027 GMT
user@DESKTOP-P8BDGDB:/etc/ssl/mycert$ sudo nano /etc/apache2/mods-available/ssl.conf
user@DESKTOP-P8BDGDB:/etc/ssl/mycert$ sudo systemctl restart apache2
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
user@DESKTOP-P8BDGDB:/etc/ssl/mycert$ sudo service apache2 restart
* Restarting Apache httpd web server apache2
[ OK ]
user@DESKTOP-P8BDGDB:/etc/ssl/mycert$ sudo service apache2 status
* apache2 is running
user@DESKTOP-P8BDGDB:/etc/ssl/mycert$
```

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